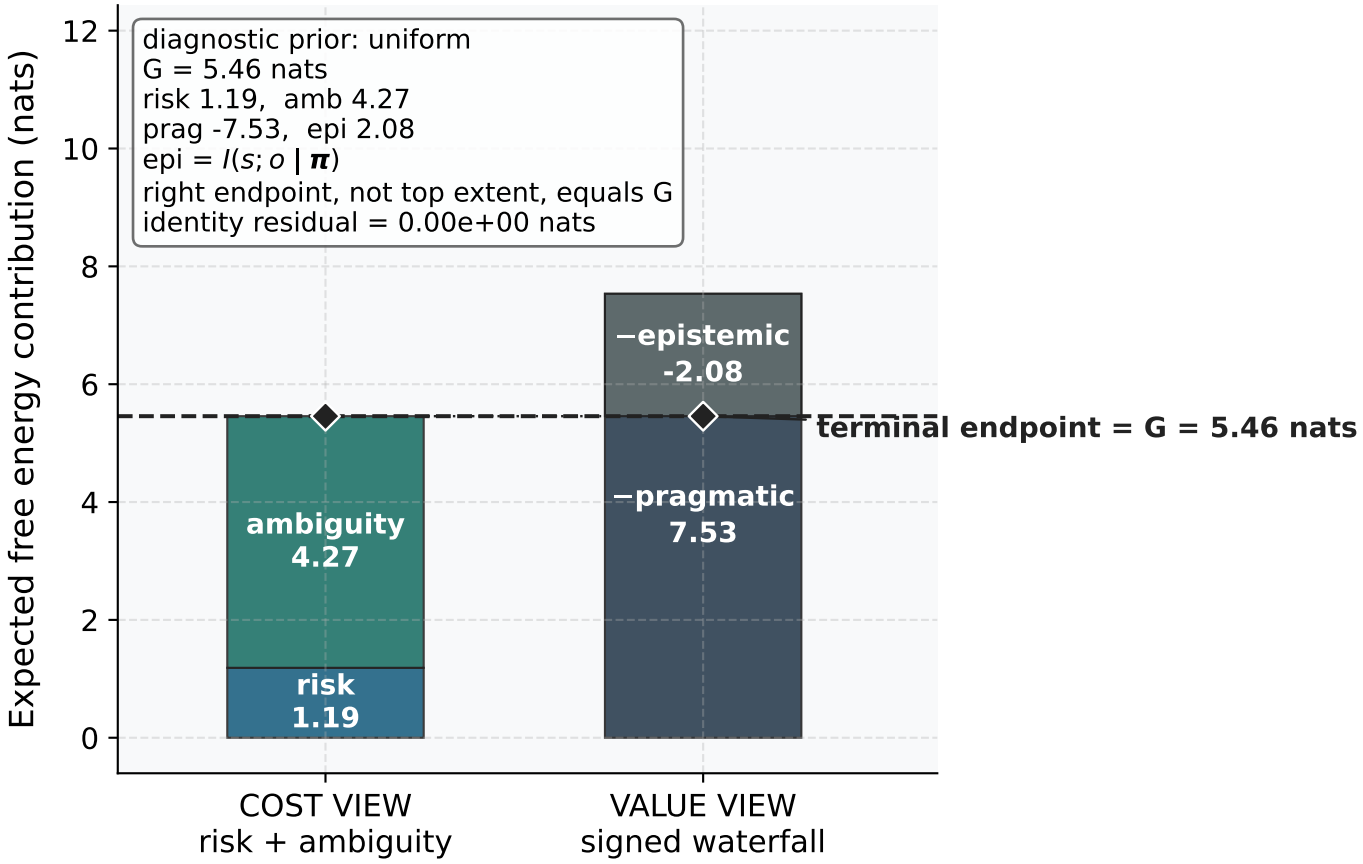
# Expected-free-energy identity: additive cost and signed value views
## Categorical specialization of Friston et al. (2024), Eq. 2




Signed waterfall: −epistemic is a negative correction;
terminal endpoint is $G$ (uniform prior makes information gain visible).

- terminal $G(\boldsymbol{\pi})$ = 5.46 nats
- risk $\mathrm{KL}(q(o|\boldsymbol{\pi}) \| p_C(o))$
- ambiguity $\mathbb{E}[H[p(o|s)]]$
- −pragmatic value
- −epistemic correction $I(s; o \mid \boldsymbol{\pi})$